Home › My Courses › Financial Markets ›
M6: Model Failure and Crises › Lesson Notes

Lesson Notes

MODULE 6 | LESSON 4

MORE THAN ONE WAY TO PICK A STOCK

Reading Time	45 minutes
Prior Knowledge	REIT
Keywords	Fundamental analysis, Macroeconomic, Microeconomic, Financial statements, Intrinsic value, Technical analysis, Trading volume, Resistance level, Support level, Momentum, Buy/sell signal, Bearish, Bullish, Lag, BRICS, Overconfidence, Anchoring, Disposition effect, Herding bias

In the last lesson, we discussed using macroeconomic data when modeling an investment, which is called fundamental analysis. There are many other types of analysis. Often, these analyses can be used together for more confident price forecasts or risk calculations.

1. Fundamental Analysis

In the last lesson, we witnessed how economics, demographics, politics, and regulation can impact an investment and should be accounted for in risk and valuation analysis. Specifically, incorporating such factors into a model represents a type of fundamental analysis because they affect the fundamental or intrinsic value of the investment. As you can already see, fundamental analysis takes into account how an investment is expected to perform based on a very broad context of macroeconomic (affecting the economy as a whole) and microeconomic (related to individual financial entities and their interactions with each other) information.

Recall from the last lesson's discussion that some real estate investment trusts (REITs) purchased much of their property at low prices in the Great Financial Crisis (a major macroeconomic event) and are now preferred by many investors due to their stable cash flows throughout the economic cycle. We also saw that we can investigate the individual residential properties owned by REITs, so we can make plausible forecasts about their ability to maintain rent revenues based on what type of renters are moving to and from a particular city and why. For example, if Facebook is expanding or cutting operations in a particular city, considering how this may affect rent revenues for a particular REIT is microeconomic analysis.

We have already seen how to value equities using dividend models in Module 2. However, that is just one of several techniques available. The most fine-tuned type of modeling is based on discounted cash flows. This method requires focused attention on the company's balance sheet and income statement and an ability to forecast the company's future based on what the company is doing and the economic environment in which it operates. Note that, for some very large companies, the environment could be more or less the entire world.

Performing a DCF is well beyond the scope of this course, but you should at least take a look at this image Example of DCF model. Notice the mix of historical data and estimated data (the blue numbers). As you can see, analysts are called upon to make a lot of choices, and to make educated guesses, they need to carefully examine the company's financial statements. This is a level of microanalysis, wherein the financial statements consist of the balance sheet that was discussed in Module 4, as well as the cash flow statement, the income statement, and the statement of equity.

Another fundamental approach to valuation uses what is called **relative valuation**. Essentially, we compare the company we are interested in to a group of carefully selected peers and use the median, or the average, or even a weighted average of the multiples of that group to value our company. The weights could be based on the market value of each of the peer companies on the total market value for the group. Alternatively, we could assign a positive or negative premium to those multiples if our company is perceived to be better or worse than the group of peers, respectively.

Here, you can find an example taken from the author's professional work about 10 years ago Relative Valuation Example. Note the many different multiples that were collected.

The first required reading (Roy 272-275) gives an overview of some basic ratios for a fundamental analysis of the financial statements using relative methods.

Some helpful notes for the first reading:

- In section 3, Projected Earnings Growth:\
    - “a high P/E ratio may make a stock look like a good buy” *should read* “a low P/E ratio may make a stock look like a good buy” (“low,” not “high”)
    - PEG = “P/E ratio 4 ÷ Annual EPS Growth” *should read* PEG = “P/E ratio ÷ Annual EPS Growth” (No “4” in the calculation formula)
- In section 5, Price to Book:\
    - “P/E Ratio = Market Price per share / Book value per share” *should read* “P/B Ratio = Market Price per share / Book value per share” (“P/B,” not “P/E”)

Where Roy indicates “Book value per share = Shareholder's Funds / Number of shares,” you should understand that “Shareholder’s Funds” refers to what we have been calling equity on the balance sheet.

The second required reading (Drakopoulou 1-8) builds on the ratios and other terminology explained in the first reading. For example, when Drakopoulou defines “relative valuation techniques, [as] the value of a stock is estimated based upon its current price relative to variables considered to be significant valuation, such as earnings, cash flow, book value, or sales” (2), you should recall the P/E, Price-to-Book, and Price-to-Sales ratios that you have already seen. You might also notice that you are already familiar with the EIC (economic, industry, and company) approach as top-down analysis. Drakopoulou expands on the basics and introduces additional analysis types, such as "Discounted Cash Flow techniques," "growth at a reasonable price (GARP) investing," and filtering investments based on performance indicators such as "Current earnings per share should be up 25%... Quarterly sales should also be up 25% or more or accelerating over prior quarters," and so on (Drakopoulou 2-3).

2. Technical Analysis

While fundamental analysis attempts to measure or forecast a stock's *intrinsic* or objective value based on macroeconomic and microeconomic information, technical analysis basically ignores these kinds of information and relies on price and trading volume (the number of shares traded in a time period) data. Technical analysis searches this data for patterns that indicate when and how the price will change. For example, a technical analyst might observe that a stock's price does not go above or below certain levels. If a stock's price increases to a certain level a few times, but does not increase above that price level, this price level is the **resistance level**. On the other hand, if a stock's price decreases to a certain level a few times, but does not decrease below that price level, this price level is the **support level**. If the stock price finally goes above the resistance level, the "bullish breakout" suggests the stock will increase significantly more (so this is a "buy signal"). If the stock price finally goes below the support level, the "bearish breakout" suggests the stock will decrease significantly more (so this is a "sell signal"). A slight variation on this incorporates moving averages of the price, instead of just the price. That is, a bullish breakout occurs when the 10-day moving average of the price exceeds the resistance level.

Another technical concept you will read about is momentum, the velocity of price changes. The simplest assessment of momentum is

Momentum = Current Price – Price "x" periods ago

For example, if Champion Breweries was trading at 2.30 NGN 10 days ago and is trading at 2.40 NGN today, the 10-day momentum is 0.10 NGN. The "10 days" is not exactly arbitrary: the period you use for the momentum calculation should be based on experience or research. And you might observe several periods of momentum: your research may indicate that the 10-day momentum is a useful indicator in some situations, but the 20-day momentum is useful in others. Depending on your holding period (e.g., how long you expect to keep a stock that you bought), you may find that 10-minute momentum is more relevant or that 100-day momentum is what you should pay attention to.

Many technical indicators are just a combination of the more basic ones, e.g., the moving average convergence divergence (MACD). This is a long name, but the actual calculation is not as intimidating; it simply tracks the difference between two moving averages, for example, the 12-day and the 26-day. When the 12-day moving average rises above the 26-day moving average, this may be a buy signal. Conversely, when the 12-day moving average falls below the 26-day moving average, this may be a sell signal.

Drakopoulou discusses these and many more technical indicators, as well as their advantages and disadvantages. For example, since a moving average is by definition the average over multiple time periods, there is some lag, or delay, before a significant price change is fully reflected. If a stock has been trading around 100 NGN for the past 10 days, the 10-day moving average price would be very close to 100. If the next day, the price drops to 90 NGN (a significant 10% drop), the 10-day moving average would now be 99, which does not by itself suggest a drastic drop. Even a 50% drop would only change the 10-day moving average to 95.

The third required reading (de Souza et al. 1-18) provides an overview of this evolution of technical analysis as well as an evaluation of technical analysis for trading stocks in the BRICS countries: Brazil, Russia, India, China, and South Africa.

We complete this section with an important remark. In the financial world, it is often said that *markets are efficient*. This means that they tend to incorporate all available information relatively fast, and opportunities to beat the market would not exist or would be present only briefly. Many professionals do not believe that, and this has created 3 different versions of market efficiency:

- **Weak Efficiency**: Today's stock prices reflect all the data of past prices, and therefore, no form of technical analysis can aid investors. Recall that technical analysis, unlike fundamental analysis, only considers past information.

- **Semi-strong Efficiency**: Public information is part of a stock's current price; therefore, investors cannot utilize either technical or fundamental analysis. As a consequence, only information not available to the public can help investors. Among the forms of non-public information, one should also include insider information, which is illegal in pretty much all jurisdictions.
- **Strong Efficiency**: All information, public and not public, is completely accounted for in current stock prices. Hence, no type of information can give an investor an advantage on the market. Basically, the market is impossible to beat.

Now, very few people believe in strong efficiency, but there is a certain amount of evidence that markets might be weakly efficient or somewhat semi-strong efficient, although there is no uniformity of opinions. A rich literature of statistical studies does exist on the topic, but that is a story for another time.

3. Behavioral Analysis

As mentioned by de Souza, technical analysis can be interpreted to “reflect the concept that price trends depend on the attitudes of individuals, i.e., the mass psychology of the crowd” (9).

Arguably, the most egregious examples where mass psychology plays a role in finance are "*bubbles*". The term **bubble**, in an economic context, generally refers to a situation where the price for an individual stock or a financial asset, or an entire sector of stocks, even an entire market, exceeds its fundamental value by a large margin.

How can this happen if investors are all rational human beings?

Well, bubbles occur because investors' demands are not driven by rationality but rather by speculation. In other words, it is not considerations about the intrinsic worth of the assets that causes prices to move.

In the beginning, demand for certain assets can be rational, but over time, valuations are out of sync with the true value of those assets. While demand lasts, the price of assets keeps rising. Some investors who were initially more cautious than the others may not have invested in those assets yet but start to feel regret about missing something potentially great, and they too buy into the bubble. This could be described as a herd effect.

This behavior causes the value of the investments to go even higher, and in some ways, it acts as a confirmation for some of the early investors that they were right to buy those assets. Thinking they are financial geniuses, they could buy even more of those assets. Some individuals who held out now start to doubt their initial decision and want in badly.

At some point, the bubble inevitably bursts because some events make it clear that the valuations are probably nothing more than wishful thinking.

However, some investors may still be reluctant to sell:

- If they were late buyers, they fear that in doing so they would lose their opportunity to make money;
- Others may stay in trying to squeeze a few more dollars out of their investments instead; and
- A few others may just be the super-optimistic types who see no problem with what is going on.

This may explain why prices at the beginning of the bubble bursting may correct slowly. Nevertheless, eventually, the bubble pops and a massive sell-off ensues.

When newspapers and media outlets in general start to doubt the real merit of certain investments or some well-known public financial figures speak against those assets or, even more so, become sellers, then the crowd at large starts to capitulate. The selling begins and the herd effect now causes a run to sell. The massive selloffs cause prices to decline even beyond their intrinsic values, but very few have the courage or the required ability to think about the actual fundamental values of those investments.

It is a fact that, in most cases, speculative bubbles are followed by equally spectacular crashes in the asset class that was affected by the bubble. This is just an example of how human behavior does not respond to cold calculations of intrinsic value. There is a lot more to this, but you will learn about it more in depth in the Portfolio Management course.

Madaan (56) explains some of the most significant findings in relation to the psychology of financial markets, namely:

- Overconfidence
- Anchoring
- Disposition effect
- Herding bias

However, there are a lot more, and as already noted a few paragraphs above, you will see an entire module on this while taking your Portfolio Management course.
At present, if you are interested, you may enjoy reading the paper by Almansour that appears among the references.

4. Sentiment Analysis

Arratia mentions that sentiment analysts “apply several ad hoc filters, as moving averages, exponential smoothers, and many other transformations to their sentiment data to concoct different indicators in order to exploit the possible dependence relation with the price or returns, or any other observable statistics” (196). This reference to prices recalls the approach taken by technical analysts. And "any other observable statistics" could include the insights gleaned from fundamental or behavior analysis.

In other words, any of these approaches can be combined with each other and/or with other approaches in novel ways. The only constraints are on your time for experimentation and testing and the availability of the required data. You may also need to take into account processing times for lengthy calculations, especially if you are trading equity in a high-frequency environment. However, the best model is not necessarily the most sophisticated or the most inclusive. On this note, this required reading about statistical arbitrage borrows from almost all of the above approaches to portfolio selection and synthesizes them into a variety of potentially profitable strategies that also significantly reduce many types of risk.

The best model depends on what you need. For example, if you want to understand a phenomenon, a simple model may be the best route. Recall the double trigger default model discussed in Lesson 1 of this module (Foote and Willen): default was predicted using a straightforward combination of options theory and some behavioral observations. Even the 5 Cs of credit analysis from Module 5 (though they do not, by themselves, constitute a fully fledged model) include intangible character alongside quantifiable capital and collateral. On the other hand, if you need the best prediction without necessarily a solid explanation for the prediction, you may find that the neural nets of deep learning are your best solution. They can often forecast very well though they are criticized for not being interpretable. This will of course be covered in a later course. This journey is still only beginning.

Very recently, sentiment analysis has emerged as a critical tool for understanding market dynamics and predicting future trends. As a consequence, the ability to analyze sentiments can provide valuable insights into the collective mood of the market, and in turn, this can enable more informed and strategic decision-making. The use of **Natural Language Processing** (NLP) has made available more sophisticated techniques compared to just a few years ago. Now, we have **deep learning**-based models available that have significantly improved sentiment analysis accuracy and reliability. However, despite these advancements, there are also challenges to sentiment analysis. For instance, financial texts, particularly news headlines, contain a lot of terminology that is very specific to certain sectors or the sentiment they convey can be very nuanced, and both elements make it quite complicated classifying those sentiments. Among the most recent tools, **ChatGPT** has emerged for its possible applications to finance given its ability to produce risk analysis through sentiment analysis, and financial institutions can enhance the accuracy and depth of sentiment analysis they rely upon.

An interesting optional reading for those who are interested in exploring this use of ChatGPT is that by Fatouros et al. listed in the references.

5. Summary: Comparing the Different Types of Analyses

To summarize, let's consider what elements can lead to a decision to buy shares in McDonald's (MACD) according to the different types of analyses we have considered in this lesson. Doing so can help you remember key features of each analysis type.

1. **Fundamental Analysis**: Operating margin has been improving, sales are stable, so the stock should do well.
2. **Technical Analysis**: We are noticing an ascending triangle pattern with the price breaking out from the upper trendline resistance. This is a buy signal, so let's buy the shares of MACD.
3. **Behavioral Analysis**: MACD is showing success at increasing purchase frequency and margins through the introduction of new products on their menus. Recently, they have been growing the number of their restaurants, incorporating new features that people seem to like. In addition, they offer home delivery, table service at some of their new restaurants, and new technology that allows for self-service. These are steps in the right direction and will make the company do well in the future because they are listening to their customers, adding new ones and retaining old ones. Their shares are likely to have a bright future.
4. **Sentiment Analysis**: The company has become more price conscious in a post-pandemic inflationary environment, offering more value meals. Its new $5- meal is gathering a lot of interest among customers who appreciate the company being on their side. Restaurant chains, including McDonald's, Burger King, and Starbucks, are promoting affordability as consumers are beginning to trade down to eat at home due to the high costs of eating out. MACD has found a way to ride the current negative trend among consumers and should do well in the near future, so the stock is becoming a buy.

6. Conclusion

In this lesson, we discussed the major types of stock analysis and mentioned some of the specific metrics, ratios, and signals they use. We also noticed that these are often related; for example, technical analysis and behavioral analysis both assume a psychological dimension to trading markets. We also saw overlap between these investment approaches: sentiment analysis uses some of the same calculations as technical analysis. Lastly, there is no reason not to combine approaches. Fundamental analysis can certainly complement technical analysis, and sentiment analysis should not ignore the intrinsic value of a stock that only fundamental analysis attempts to calculate.

The first lesson of Module 7 will also consider valuation approaches with an expanded set of assets that includes not only bonds and options but also exchange-traded funds (ETFs).

References

- Almansour, Bashar Yaser, et al. "Behavioral Finance Factors and Investment Decisions: A Mediating Role of Risk Perception." *Cogent Economics & Finance*, vol. 11, no. 2, 2023, pp. 1-20. https://doi.org/10.1080/23322039.2023.2239032.
- Arratia, Argimiro, et al. “Sentiment Analysis of Financial News: Mechanics and Statistics.” *Data Science for Economics and Finance*, edited by Sergio Consoli, et al., Springer, 2021, pp. 195-216. https://doi.org/10.1007/978-3-030-66891-4_9.
- de Souza, Matheus Jose Silva, et al. “Examination of the Profitability of Technical Analysis Based on Moving Average Strategies in BRICS.” *Financial Innovation*, vol. 4, no. 3, 2018, pp. 1-18. https://doi.org/10.1186/s40854-018-0087-z.
- Drakopoulou, Veliota. “A Review of Fundamental and Technical Stock Analysis Techniques.” *Journal of Stock & Forex Trading*, vol. 5, no. 1, 2015, pp. 1-8. https://www.longdom.org/open-access/a-review-of-fundamental-and-technical-stock-analysis-techniques-2168-9458-1000163.pdf.
- Fatouros, Gergios, et al. "Transforming Sentiment Analysis in the Financial Domain with ChatGPT." *Machine Learning with Applications*, vol. 14, no. 15, December 2023. https://doi.org/10.1016/j.mlwa.2023.100508.
- Lazzarino, Marco, et al. "What is Statistical Arbitrage?" *Theoretical Economics Letters*, vol. 8, 2018, pp. 888-908. https://www.scirp.org/journal/paperinformation?paperid=83611.
- Madaan, Geetika, and Sanjeet Singh. “An Analysis of Behavioral Biases in Investment Decision-Making.” *International Journal of Financial Research*, vol. 10, no. 55, 2019. https://doi.org/10.5430/ijfr.v10n4p55.
- Roy, Sreemoyee Guha. “Equity Research: Fundamental and Technical Analysis.” *International Journal of Science and Research*, vol. 4, no. 9, Sept. 2015, pp. 272-79. https://www.ijsr.net/archive/v4i9/SUB157950.pdf.